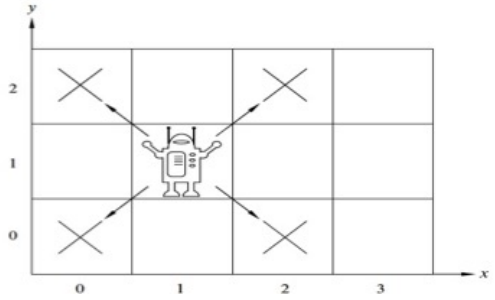
# Structural induction example robot grid




**Theorem**: A robot on an infinite 2-dimensional integer grid starts at $(0, 0)$ and at each step moves to diagonally adjacent grid point. This robot can / cannot (*circle one*) reach $(1, 0)$.

**Definition** The set of positions the robot can visit $Pos$ is defined by:

Basis Step: $(0, 0) \in Pos$
Recursive Step: If $(x, y) \in Pos$, then

are also in $Pos$

*Example elements of Pos are*:

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**Lemma**: $\forall(x, y) \in Pos$ ($x + y$ is an even integer )

*Why are we calling this a lemma?*

Proof of theorem using lemma: To show is $(1, 0) \notin Pos$. Rewriting the lemma to explicitly restrict the domain of the universal, we have $\forall(x, y)$ ( $(x, y) \in Pos \rightarrow (x+y$ is an even integer) ). Since the universal is true, ( $(1, 0) \in Pos \rightarrow (1 + 0$ is an even integer) ) is a true statement. Evaluating the conclusion of this conditional statement: By definition of long division, since $1 = 0 \cdot 2 + 1$ (where $0 \in \mathbb{Z}$ and $1 \in \mathbb{Z}$ and $0 \leq 1 < 2$ mean that 0 is the quotient and 1 is the remainder), $1$ **mod** $2 = 1$ which is not 0 so the conclusion is false. A true conditional with a false conclusion must have a false hypothesis: $(1, 0) \notin Pos$, QED. □

Proof of lemma by structural induction:

**Basis Step**:

**Recursive Step**: Consider arbitrary $(x, y) \in Pos$. To show is:

($x + y$ is an even integer) → (sum of coordinates of next position is even integer)

Assume **as the induction hypothesis, IH** that:

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# Fundamental theorem proof

**Theorem**: Every positive integer *greater than 1* is a product of (one or more) primes.

**Before we prove, let's try some examples**:

$20 =$

$100 =$

$5 =$

**Proof by strong induction**, with $b = 2$ and $j = 0$.

**Basis step**: WTS property is true about 2.

Since 2 is itself prime, it is already written as a product of (one) prime.

**Recursive step**: Consider an arbitrary integer $n \geq 2$. Assume (as the strong induction hypothesis, IH) that the property is true about each of $2, \ldots, n$. WTS that the property is true about $n + 1$: We want to show that $n + 1$ can be written as a product of primes. Notice that $n + 1$ is itself prime or it is composite.

*Case 1*: assume $n + 1$ is prime and then immediately it is written as a product of (one) prime so we are done.

*Case 2*: assume that $n + 1$ is composite so there are integers $x$ and $y$ where $n + 1 = xy$ and each of them is between 2 and $n$ (inclusive). Therefore, the induction hypothesis applies to each of $x$ and $y$ so each of these factors of $n + 1$ can be written as a product of primes. Multiplying these products together, we get a product of primes that gives $n + 1$, as required.

Since both cases give the necessary conclusion, the proof by cases for the recursive step is complete.

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# Strong induction making change proof idea

Suppose we had postage stamps worth 5 cents and 3 cents. Which number of cents can we form using these stamps? In other words, which postage can we pay?

11?

15?

4?

$$CanPay(0) \wedge \neg CanPay(1) \wedge \neg CanPay(2) \wedge$$
$$CanPay(3) \wedge \neg CanPay(4) \wedge CanPay(5) \wedge CanPay(6)$$
$$\neg CanPay(7) \wedge \forall n \in \mathbb{Z}^{\geq 8} CanPay(n)$$

where the predicate $CanPay$ with domain $\mathbb{N}$ is

$$CanPay(n) = \exists x \in \mathbb{N} \exists y \in \mathbb{N}(5x + 3y = n)$$

**Proof** (idea): First, explicitly give witnesses or general arguments for postages between 0 and 7. To prove the universal claim, we can use mathematical induction or strong induction.

*Approach 1, mathematical induction*: if we have stamps that add up to $n$ cents, need to use them (and others) to give $n + 1$ cents. How do we get 1 cent with just 3-cent and 5-cent stamps?
Either take away a 5-cent stamps and add two 3-cent stamps,
or take away three 3-cent stamps and add two 5-cent stamps.
The details of this proof by mathematical induction are making sure we have enough stamps to use one of these approaches.

*Approach 2, strong induction*: assuming we know how to make postage for **all** smaller values (greater than or equal to 8), when we need to make $n+1$ cents, add one 3 cent stamp to however we make $(n + 1) - 3$ cents.
The details of this proof by strong induction are making sure we stay in the domain of the universal when applying the induction hypothesis.

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# Strong induction nim

Consider the following game: two players start with two (equal) piles of jellybeans in front of them. They take turns removing any positive integer number of jellybeans at a time from one of two piles in front of them in turns.

The player who removes the last jellybean wins the game.

Which player (if any) has a strategy to guarantee to win the game?

Try out some games, starting with 1 jellybean in each pile, then 2 jellybeans in each pile, then 3 jellybeans in each pile. Who wins in each game?

Notice that reasoning about the strategy for the 1 jellybean game is easier than about the strategy for the 2 jellybean game.

*Formulate a winning strategy by working to transform the game to a simpler one we know we can win.*

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*Player 2's Strategy*: Take the same number of jellybeans that Player 1 did, but from the opposite pile.

*Why is this a good idea*: If Player 2 plays this strategy, at the next turn Player 1 faces a game with the same setup as the original, just with fewer jellybeans in the two piles. Then Player 2 can keep playing this strategy to win.

**Claim**: Player 2's strategy guarantees they will win the game.

**Proof**: By strong induction, we will prove that for all positive integers $n$, Player 2's strategy guarantees a win in the game that starts with $n$ jellybeans in each pile.

**Basis step**: WTS Player 2's strategy guarantees a win when each pile starts with 1 jellybean.

In this case, Player 1 has to take the jellybean from one of the piles (because they can't take from both piles at once). Following the strategy, Player 2 takes the jellybean from the other pile, and wins because this is the last jellybean.

**Recursive step**: Let $n$ be a positive integer. As the strong induction hypothesis, assume that Player 2's strategy guarantees a win in the games where there are $1, 2, \ldots, n$ many jellybeans in each pile at the start of the game.

WTS that Player 2's strategy guarantees a win in the game where there are $n+1$ jellybeans in each pile at the start of the game.

In this game, the first move has Player 1 take some number, call it $c$ (where $1 \leq c \leq n+1$), of jellybeans from one of the piles. Playing according to their strategy, Player 2 then takes the same number of jellybeans from the other pile.

Notice that $(c = n+1) \vee (c \leq n)$.

*Case 1*: Assume $c = n+1$, then in their first move, Player 2 wins because they take all of the second pile, which includes the last jellybean.

*Case 2*: Assume $c \leq n$. Then after Player 2's first move, the two piles have an equal number of jellybeans. The number of jellybeans in each pile is

$$(n+1) - c$$

and, since $1 \leq c \leq n$, this number is between 1 and $n$. Thus, at this stage of the game, the game appears identical to a new game where the two piles have an equal number of jellybeans between 1 and $n$. Thus, the strong induction hypothesis applies, and Player 2's strategy guarantees they win.